

Nirav Diwan

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Education

Aug 2024 –	University of Illinois Urbana-Champaign (UIUC) Ph.D. in Computer Science	Champaign, US
Aug 2022 Aug 2024	University of Illinois Urbana-Champaign (UIUC) M.S in Computer Science (thesis-track)	Champaign, US
Aug 2017 Jun 2021	Indraprastha Institute of Information Technology, Delhi (IIITD) Bachelor in Technology (B.Tech) in Computer Science & Engineering (CSE) 🏆 Dean's Award for Academic Excellence	Delhi, India

Publications

PR = In Preparation, S=In Submission, C=Conference, W=Workshop, P=Poster/Demo, J=Journal, *Equal Contribution

- [S.1] **SAJE: Turning Model Specifications against the Model** [🔗] [📄]
Nirav Diwan, Xiangwen Wang, Varun Chandrasekaran, Gang Wang
The 36th USENIX Security Symposium, 2027 [USENIX Security 2027]
▶ We use the Model Specifications (e.g OpenAI Model Spec) for jailbreaking OpenAI's GPT-OSS-20B, beating baselines by 30 pp.
- [C.7] **CoT-Guard: Small Models for Strong Practical Monitoring** [🔗] [📄]
Nirav Diwan*, Han Wang*, Berkcan Kapusuzoglu, Ramin Moradi, Supriyo Chakraborty, Giri Iyengar, Sambit Sahu, Huan Zhang, Gang Wang
The 40th Annual Conference on Neural Information Processing Systems, 2026 [NeurIPS 2026]
The 1st Workshop on AdvML-Frontiers × CoTMA (Oral), COLM 2026 [AdvML × CoTMA (Oral) @ COLM 2026]
▶ We propose a new CoT monitoring threat model, and use RL to train small models (~4B) to outperform large models (GPT-5.4).
- [W.3] **Extractable Memorization from a Differentially Private Pre-trained LLM** [🔗]
Nirav Diwan, Gang Wang, Daniel Alabi
Theory and Practice of Differential Privacy (TPDP) Workshop, 2026 [TPDP 2026]
▶ We extract Personally Identifiable Information (PII) from VaultGemma — the first LLM trained through DP-SGD.
- [W.2] **Beyond BeautifulSoup: Benchmarking LLM-Powered Web Scraping for Everyday Users** [🔗]
Arth Bhardwaj, Nirav Diwan, Gang Wang
2nd Artificial Intelligence for Cyber Security (AICS) Workshop (Oral) @ AAI, 2026 [AICS @ AAI 2026]
▶ My awesome mentee (Arth) developed a benchmark to check if LLMs can scrape websites with minimal prompting!
- [C.6] **PurpCode: Reasoning for Safer Code Generation** [🔗] [📄]
Jiawei Liu*, Nirav Diwan*, Zhe Wang*, Haoyu Zhai, Xiaona Zhou, Kiet A. Nguyen, Tianjiao Yu, Muntasir Wahed, Yinlin Deng, Hadjer Benkraouda, Yuxiang Wei, Lingming Zhang, Ismini Lourentzou, Gang Wang
39th Annual Conference on Neural Information Processing Systems [NeurIPS 2025]
🏆 1st place in the Amazon Nova AI Challenge (\$250,000)
▶ Developed a reasoning model for secure code generation using Deliberative Alignment.
- [W.1] **Clear Preferences Leave Traces: Reference Model-Guided Sampling for Preference Learning** [🔗]
Nirav Diwan, Tolga Ergen, Dongsub Shim, Honglak Lee
1st Workshop on Preparing Good Data for Generative AI: Challenges and Approaches, 2025 [Good Data Workshop @ AAI 2025]
▶ Fine-tuned models identify quality preference pairs for alignment *without* knowing which response is better.
- [C.5] **MOCHA: Are Code Language Models Robust Against Multi-Turn Malicious Coding Prompts?** [🔗]
Muntasir Wahed, Xiaona Zhou, Kiet A. Nguyen, Tianjiao Yu, Nirav Diwan, Gang Wang, Dilek Hakkani-Tur, Ismini Lourentzou
30th Empirical Methods in Natural Language Processing (Findings), 2025 [EMNLP 2025]
▶ A multi-turn benchmark for jailbreaking code models
- [C.4] **You Can't Judge a Binary by Its Header: Data-Code Separation for Non-Standard ARM Binaries using Pseudo Labels** [🔗]
Hadjer Benkraouda, Nirav Diwan, Gang Wang
46th IEEE Symposium on Security and Privacy, 2025 [IEEE S&P 2025]
▶ We propose a novel way to separate data and code in non-standard ARM binaries using pseudo labels.
- [C.3] **It Doesn't Look Like Anything to Me: Using Diffusion Models to Subvert Visual Phishing Detectors** [🔗] [📄]
Qingying Hao, Nirav Diwan, Ying Yuan, Mauro Conti, Giovanni Apruzzese, Gang Wang
33rd USENIX Security Symposium, 2024 [USENIX Security 2024]
▶ Used diffusion models to attack phishing detectors, validated across 100+ brands in white-box/black-box settings.

[C.2]

Weakening the Inner Strength: Spotting Core Collusive Users in YouTube Blackmarket Networks [🔗] [📄]

Hridoy Sankar Dutta*, [Nirav Diwan](#)*, Tanmoy Chakraborty

16th International AAAI Conference on Web and Social Media, 2022

[ICWSM 2022]

➤ Investigated the collusive blackmarket on YouTube using graphs to identify the most influential blackmarket users.

[C.1]

Fingerprinting Fine-tuned Language Models in the Wild [🔗] [📄]

[Nirav Diwan](#), Tanmoy Chakraborty, Zubair Shafiq

59th Annual Meeting of the Association for Computational Linguistics (Findings), 2021

[ACL 2021]

🏆 **Dean's Thesis Appreciation Award**

➤ Developed an LLM-based classifier to attribute AI-generated text to its fine-tuned language model across 100+ classes.

[J.1]

RecipeDB: A Resource for Exploring Recipes [🔗] [📄]

Devansh Batra*, [Nirav Diwan](#)*, Utkarsh Upadhyay*, Jushaan Singh Kalra*, Tript Sharma*, Aman Kumar Sharma*, Dheeraj Khanna*, Jaspreet Singh Marwah*, Srilakshmi Kalathil*, Navjot Singh*, Rudraksh Tuwani*, Ganesh Bagler*

Database: The Journal of Biological Databases and Curation, Oxford University Press, 2020

[Database 2020]

🌐 **Press:** [Times of India](#), [The National](#), [Nature India](#)

➤ Developed a worldwide database of recipes with over 100,000 recipes from 50+ countries.

Awards

Amazon Nova AI Challenge (Winner): Won 1st place as team co-lead in the [Amazon Nova AI Challenge](#) winning a cash prize of \$250K.

Amazon Nova AI Grant: Awarded grants worth \$250,000 + \$1.5 Million AWS Credits.

Catalyzing Advocacy in Science and Engineering (CASE) Workshop hosted by the AAAS (American Association for the Advancement of Science) 2024: Selected to represent the CS department from UIUC. Informed legislators about AI progress.

Dean's Thesis Appreciation Award, IIITD: For outstanding thesis research from Dean of Academic Affairs.

Dean's Award for Academic Excellence, IIITD: For exceptional academic performance for the academic years 2019-20.

IIT-JEE 2017: Ranked 1884/1.5 million (top 0.15%) in IIT-JEE Mains and 4186/200,000 (top 2%) in IIT-JEE Advanced.

Recent Industrial Experience

1. LG AI Research | Bilingual LLM Team [🌐]

Ann Arbor, US

Applied Science Intern | Area: Natural Language Processing (NLP)

2024

➤ Improved alignment of LLMs for single-turn and multi-turn settings. Results published in the **AAAI Good Data Workshop**.

➤ Developed sampling methods for alignment methods (e.g DPO) for Coding, Math, and Reasoning – increasing MT-Bench scores by **+1.0**.
2. Ema Inc. | Generative AI Team [🌐]

Remote, US

Applied Science Intern | Area: Natural Language Processing (NLP)

2023

➤ Led the large-scale deployment of a conversational AI agent for translating natural language queries into SQL (Text2SQL).

➤ Built an eval pipeline and benchmark dataset of 3,000+ real-world Text2SQL examples for robust evaluation.

Selected Research Experience

1. University of Illinois Urbana-Champaign (UIUC) | Security & Privacy Group [🌐]

Champaign, US

Graduate Researcher | Advisor: [Prof. Gang Wang](#) | Areas: Machine Learning (ML) and Security & Privacy (S&P)

2022 - Present

➤ Identifying and protecting against practical underperformance, attacks, and misuse of AI models.

➤ Published projects at **NeurIPS**, **IEEE S&P** and **USENIX Security**.
2. IIT Delhi | Laboratory for Computational Social Systems (LCS2) [🌐]

Delhi, India

Undergraduate Researcher | Advisor(s): [Prof. Zubair Shafiq](#), [Prof. Tanmoy Chakraborty](#) | Areas: NLP, S&P and Graphs

2019 - 2021

➤ Identified malicious actors online using machine learning techniques.

➤ Published projects at **ACL** and **ICWSM**.

Academic Service

Reviewer	NeurIPS 2026, MemFM + TGAIR ICML 2026, ICLR'22 (Student Reviewer)
Volunteer	ACM FAccT'23, AAAI ICSWM'21, AAAI ICWSM'22, ACL'21